

AIDS-1 Unique Questions

Module I: Introduction to AI & Intelligent Agents

#	Question	Marks	Month & Year
1	What is a rational agent? Explain with diagram.	5	Dec 2023
2	Explain PEAS and also write down the PEAS representations for Medical diagnosis system.	10	Dec 2023
3	State PEAS of automated taxi driver.	5	May 2023
4	Explain learning agents with example.	5	Dec 2024
5	What are the different types of environments? Give examples. Explain the vacuum world problem with its environment.	10	Dec 2024
6	Differentiate between Simple Reflex Agent and Model-Based Reflex Agent.	5	Dec 2025
7	Discuss Problem Formulation in AI with an example.	5	Dec 2025
8	Formulate the problem for 8 queens.	5	May 2024
9	Explain utility based agent architecture with diagram.	5	May 2024
10	Explain PEAS descriptors. Provide PEAS descriptors for Part Picking robot.	5	May 2025
11	Explain goal based agents with example.	5	May 2025
12	Explain state space based problem formulation. Formulate 8 Puzzle and N Queens problem.	10	May 2025
13	What are the different types of environments? Give examples. Explain the vacuum world problem with its environment.	10	May 2025

Module II: Search Techniques

#	Question	Marks	Month & Year
1	What are heuristic functions? Where are they used?	5	Dec 2023

#	Question	Marks	Month & Year
2	Explain uniform cost search and best first search in detail with examples and compare.	10	Dec 2023
3	Can 1 liter water be measured using 10 liter and 4 liter jug? Justify.	10	May 2023
4	State A* algorithm and explain with example how A* searching algorithm helps in finding the goal with optimal path.	10	May 2023
5	Can min-max be used for team games? Draw sample trees for 2 and 3 teams.	10	May 2023
6	Write Short Note on: Alpha Beta Pruning	5	Dec 2024
7	Write Short Note on: A* algorithm	5	Dec 2024
8	Define heuristic search. Give two examples of heuristic functions.	5	Dec 2025
9	Explain the working of the Hill Climbing algorithm. What are issues in Hill Climbing?	10	Dec 2025
10	Represent given (5*5) grid (S start, G goal, # obstacle) as state space search problem. Apply Hill Climbing Search. Does the algorithm get stuck in Local Minima?	10	May 2024
11	Describe any four uninformed search strategies. Compare them with time complexity, space complexity, optimality and completeness.	10	May 2024
12	Compare Informed and Uninformed Search algorithms. Explain working of A* algorithm with an example. Explain which category of search algorithms it belongs to.	10	May 2025
13	Explain the working of the Hill climbing algorithm. What are issues in Hill climbing?	10	May 2025
14	Write Short Note on: Min Max Algorithm	5	May 2025

Module III: Knowledge Representation using First Order Logic

#	Question	Marks	Month & Year
1	What is Unification? Give example.	5	Dec 2023
2	What is Skolemization? Explain Skolem constant and Skolem function.	5	Dec 2023
3	The law says it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold to it by Colonel West, who is American. Prove that Col. West is a criminal!	10	Dec 2023
4	1. Marcus was a man. 2. Marcus was a Pompeian. 3. All Pompeians were Romans. 4. Caesar was a ruler. 5. All Pompeians were either loyal to Caesar or hated him. 6. Every one is loyal to someone. 7. People only try to assassinate rulers they are not loyal to. 8. Marcus tried to assassinate Caesar. Was Marcus loyal to Caesar? Solve using resolution.	10	May 2023
5	What are the rules of conversion from predicate to CNF? Explain each rule with proper example.	10	May 2023
6	Write Short Note on: CNF	5	Dec 2024
7	Write Short Note on: Skolemization	5	Dec 2024
8	Write Short Note on: Rules in Propositional Logic	5	Dec 2024
9	Describe the process of knowledge representation using First Order Logic.	5	Dec 2025
10	Discuss Partial Order, Hierarchical, and Conditional Planning.	10	Dec 2025
11	What is the difference between unification and Skolemization?	5	May 2024
12	Explain working of Forward and Backward Chaining algorithms.	5	May 2024
13	Write Short Note on: Forward chaining based proofs	5	May 2025

Module IV: Introduction to Data Science

#	Question	Marks	Month & Year
1	Explain various stages in the Data analytics Lifecycle.	10	Dec 2023
2	In detail, explain steps in the Data Science Project.	10	May 2023
3	Differentiate between data scientists, big data professionals and data analysts.	10	Dec 2024
4	Describe in detail the stages in the Data Science Lifecycle.	10	Dec 2025
5	Compare and contrast Data Science, Business Analytics, and Big Data with examples.	10	Dec 2025
6	What do you mean by data analytics? What are the different types of data analytics?	10	May 2024
7	Describe the architecture of ML application. Explain with a diagram.	10	May 2024
8	Differentiate between data scientists, big data professionals and data analysts.	10	May 2025
9	Describe steps for developing ML applications with a labeled diagram.	10	May 2025

Module V: Exploratory Data Analysis (EDA)

#	Question	Marks	Month & Year
1	Explain what role is played by Correlation and Covariance in EDA?	5	Dec 2023
2	What do you mean by covariance and correlation? Explain the range of coefficients of correlation and covariance. Calculate COV and CORRCOV for given data (Observed Value1, Observed Value2). How do you interpret these values? (Data table provided)	10	Dec 2023
3	With respect to Quantitative data analysis explain: i. Measure of central tendencies. ii. Measure of spread iii. Skewness and Kurtosis	10	May 2023
4	Compare barplot and histogram.	5	Dec 2024
5	What do you mean by covariance and correlation? Explain the range of coefficients of correlation and covariance. Calculate COV and CORRCOV for given data (Observed Value1, Observed Value2). (Data table provided)	10	Dec 2024

#	Question	Marks	Month & Year
6	Compare Z-Test, T-Test and ANOVA in detail.	10	Dec 2024
7	What are the different univariate plots in EDA? Explain them in detail.	10	Dec 2024
8	Compare and contrast univariate and multivariate EDA with suitable examples.	10	Dec 2025
9	Explain the importance of graphical methods in EDA.	10	Dec 2025
10	What do you mean by covariance and correlation? Explain their interpretation and the range of coefficients. Calculate Covariance and Correlation Coefficient for given X and Y data. Interpret results.	10	Dec 2025
11	What are the different univariate plots in EDA? Explain them in detail.	10	May 2024
12	Perform t-test on given Tea/Coffee customer preference data (5 trials). Comment if means are same. State Null/Alternate hypothesis.	10	May 2024
13	What are the different univariate plots in EDA? Explain them in detail.	10	May 2025
14	For given data (Observed Value 1 & 2), draw scatter plot. Comment on relationship. Find Coefficient of correlation. Does statistic confirm observation?	10	May 2025
15	Write Short Note on: ANOVA	5	May 2025
16	Write Short Note on: Non graphical EDA	5	May 2025
17	Write Short Note on: Box Plot	5	May 2025
18	Perform one way ANOVA F Test on house price prediction data (Models A, B, C, 5 trials) and comment if mean prices are same ($\alpha=0.05$).	10	May 2023

Module VI: Introduction to Machine Learning

#	Question	Marks	Month & Year
1	Write comparison between Business Intelligence and Data Science.	5	Dec 2023
2	Write in detail issues in machine learning.	10	Dec 2023
3	Elaborate in detail the steps in developing a Machine Learning application with architectural diagram.	10	Dec 2023
4	Explain SVM in detail.	10	Dec 2023
5	Compare Linear Regression vs Logistic Regression with suitable diagrams and formulas.	10	May 2023
6	What are the different types of Machine Learning algorithms? Give example of each category.	10	May 2023
7	What are the different issues in ML algorithms?	5	Dec 2024
8	Define and state effects of overfitting and underfitting.	5	Dec 2024
9	What is SVM? Explain its significance in ML and compare it with logistic regression.	10	Dec 2024
10	What is linear regression? Explain its significance in ML and compare it with logistic regression.	10	Dec 2024
11	Describe steps for developing ML applications with a labeled diagram.	10	Dec 2024
12	What are Supervised Machine Learning techniques? Explain Logistic Regression technique. What makes it a Supervised learning technique?	10	Dec 2025
13	Explain the basic steps in developing a Machine Learning application.	10	Dec 2025
14	Define Unsupervised Machine Learning techniques. Discuss K-Means and Hierarchical Clustering in detail.	10	Dec 2025
15	Define overfitting and underfitting in detail with a diagram.	5	May 2024
16	Write note on following supervised learning techniques: a) SVM b) ID3	10	May 2024
17	What is linear regression? Explain its significance in ML. Compare it with logistic regression.	10	May 2024
18	What are the different issues in ML algorithms?	10	May 2024
19	What are the different issues in ML algorithms?	5	May 2025
20	Define and state effects of overfitting and underfitting.	5	May 2025
21	Describe steps for developing ML applications with a labeled diagram.	10	May 2025